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Assignment: Count Words with MapReduce

✓ Passed and verified · 100/100 points

Deadline Pass this assignment by October 18, 11:59 PM PDT.

Instructions (</learn/intro-to-big-data/programming/T69HR/count-words-with-mapreduce>)

[My submission \(/learn/intro-to-big-data/programming/T69HR/count-words-with-mapreduce/submission\)](/learn/intro-to-big-data/programming/T69HR/count-words-with-mapreduce/submission)

[Discussions \(/learn/intro-to-big-data/programming/T69HR/count-words-with-mapreduce/discussions\)](/learn/intro-to-big-data/programming/T69HR/count-words-with-mapreduce/discussions)

Motivation: Although you have already run your first Hadoop program with Hello World, we want you to be able to experience the kinds of analysis Hadoop is used for in the real world. We'll do this by having you copy in some code you may not understand just yet, but which gives you a preview of (at least the basics) of what we will learn to do in Hadoop in the next course!

How you Will Submit: As you follow the remaining instructions on this page, you will be directed to run a word counting program in Hadoop. After you finish, you will go on to the 'My Submission' tab where you will be asked to upload a file containing the output of running this program. We know it's simple, but we want to give you some points for completing this important task of running a program in Hadoop!

Copy in MapReduce Code

First, be sure you are logged in to Coursera and are viewing these instructions using a web browser within the Cloudera Quickstart VM. If not, you may have difficulty copying and pasting.

1. Open a Terminal (Right-click on Desktop or click Terminal icon in the top toolbar). Follow the steps below, copying and pasting or typing the code below to execute a simple word count example.

2. Change Directory to the Hadoop library directory (type):

```
cd /usr/lib/hadoop-mapreduce/
```

3. Execute the Hadoop jar command to run the WordCount example (type):

Help Center

```
hadoop jar hadoop-mapreduce-examples.jar wordcount
```

4. The wordcount example will output the warning that it needs input and output parameters (you should see this):

```
Usage: wordcount <in> <out>
```

5. Let's create a couple of text files with a few words in it for testing, or use a log file (type both commands, one at a time):

```
echo "Hello world in HDFS" > /home/cloudera/testfile1
```

```
echo "Hadoop word count example in HDFS" > /home/cloudera/testfile2
```

6. Create directory for the input files on the HDFS file system (type):

```
hdfs dfs -mkdir /user/cloudera/input
```

7. Copy the files from local filesystem to the HDFS filesystem (type both commands, one at a time):

```
hdfs dfs -put /home/cloudera/testfile1 /user/cloudera/input
```

```
hdfs dfs -put /home/cloudera/testfile2 /user/cloudera/input
```

Run the Program

8. Run the Hadoop WordCount example with the input and output specified (type):

```
hadoop jar hadoop-mapreduce-examples.jar wordcount /user/cloudera/input /user/cloudera/output
```

9. Hadoop prints out a large amount of logging information. After completion view the output directory (type):

```
hdfs dfs -ls /user/cloudera/output
```

10. Check the output file to see the results (type):

```
hdfs dfs -cat /user/cloudera/output/part-r-00000
```

11. Put the output in a location, in this case your cloudera user's home directory, where you can upload it for your submission (type):

```
hdfs dfs -get /user/cloudera/output/part-r-00000 /home/cloudera/wordcount.txt
```

After you finish with the last command in the Step 11, you will go on to the 'My Submission' tab where you will be asked to upload the wordcount.txt file which contains the output of running this program.

How to submit

When you're ready to submit, you can upload files for each part of the assignment on the "My submission" tab.

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